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Thyroid lobectomy is not sufficient for T2 papillary thyroid cancers

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ABSTRACT

Background: Histologic subtypes of papillary thyroid cancer affect prognosis. The objective of this study was to examine whether survival is affected by extent of surgery for conventional versus follicular-variant papillary thyroid cancer when stratified by tumor size.

Methods: Using the National Cancer Data Base, we evaluated 33,816 adults undergoing surgery for papillary thyroid cancer from 2004 to 2008 for 1.0-3.9 cm tumors and clinically negative lymph nodes. Conventional and follicular-variant papillary thyroid cancers were divided into separate groups. Cox regression models stratified by tumor size were used to determine if extent of surgery affected overall survival.

Results: A total of 30,981 patients had total thyroidectomy and 2,835 had thyroid lobectomy; 22,899 patients had conventional papillary thyroid cancer and 10,918 had follicular-variant papillary thyroid cancer. On unadjusted KM analysis, total thyroidectomy was associated with improved survival for conventional ($P = 0.02$) but not for follicular-variant papillary thyroid cancer patients ($P = 0.42$). For conventional papillary thyroid cancer, adjusted analysis showed total thyroidectomy was associated with improved survival for 2.0-3.9 cm tumors ($P = 0.03$) but not for 1.0-1.9 cm tumors ($P = 0.16$). For follicular-variant, lobectomy and total thyroidectomy had equivalent survival for 1.0-1.9 cm ($P = 0.45$) and 2.0-3.9 cm ($P = 0.88$) tumors.

Conclusion: Tumor size, histologic subtype, and surgical therapy are important factors in papillary thyroid cancer survival. Total thyroidectomy was associated with improved survival in patients with 2.0-3.9 cm conventional papillary thyroid cancer, and should be considered for 2.0-3.9 cm papillary thyroid cancers when preoperative molecular analysis is not used to distinguish conventional from follicular-variant.

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Thyroid cancer is the most rapidly increasing cancer in the United States. From 2003–2012, the rate of thyroid cancer increased by an average of 5.1% per year. The American Cancer Society projected an incidence of 64,300 cases of thyroid cancer in 2016 in the United States. In addition, thyroid cancer is expected to be the fifth leading cause of new cancer diagnoses among female patients in the United States and the second leading cause among Asian American, Native Hawaiian, and Pacific Islander females in the United States for 2016.[1] Papillary thyroid cancer (PTC) is the most common type of thyroid cancer, representing an estimated 90% of all thyroid cancers.[2] PTC has several variants, and the prognosis differs among the various histologic subtypes.[3]

The 2015 American Thyroid Association (ATA) guidelines suggest that for patients with low risk papillary carcinomas >1 cm and <4 cm without extrathyroidal extension, and without clinical evidence of lymph node metastases (cN0), thyroid lobectomy alone may be sufficient initial treatment.[4] This recommendation was supported by a National Cancer Data Base (NCDB) analysis by Adam et al of 61,775 patients from 1998–2006 with PTC tumors between 1.0 and 4.0 cm who were found to have a similar overall survival regardless of whether they underwent lobectomy or total thyroidectomy.[5] These findings contradicted a previous NCDB study by Bilimoria et al of 52,173 PTC patients in whom total thyroidectomy for tumors >1 cm was found to provide an overall survival advantage.[6] Several factors have been proposed to explain the conflicting results, specifically the lack of data on comorbidity, multifocality, and extrathyroidal extension in the Bilimoria et al study.

Our objective was to determine if any other factors could account for the differences seen in these studies. Specifically, we sought to examine whether different histologic subtypes of PTC have a different overall survival based on tumor size and extent of surgery using a recent patient cohort (2004–2008), while accounting for the aforementioned variables.

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Methods

The NCDB was chosen for review so that a comparison could be made with previous similar studies addressing this topic. The NCDB is a clinical oncology database sponsored by the American College of Surgeons and American Cancer Society and comprises hospital registry data collected from more than 1,500 Commission on Cancer accredited facilities. The NCDB contains nearly 30 million historical records and ≈70% of newly diagnosed cancer patients nationwide.[7]

Using the 2013 NCDB participant user file, all patients who underwent thyroid surgery for thyroid cancer from 2004 to 2008 were identified. The earliest year of 2004 was chosen because data on multifocality, extrathyroidal extension, and comorbidity index were only available after 2004, and their effects on survival would need to be accounted for in the regression model. The final year 2008 was chosen to provide a minimum of 5 years of survival data. Using the *International Classification of Disease for Oncology, Third Edition* (ICD-O-3), 7 subtypes of PTC were identified: 8050/3 (papillary carcinoma, not otherwise specified [NOS]), 8260/3 (papillary adenocarcinoma, NOS), 8340/3 (papillary carcinoma, follicular variant [FVPTC]), 8341/3 (papillary microcarcinoma), 8342/3 (papillary carcinoma, oxyphilic cell), 8343/3 (papillary carcinoma, encapsulated), and 8344/3 (papillary carcinoma, columnar cell).[8] More than 90% of PTC patients in the NCDB are reported as papillary carcinoma, NOS (8050/3); papillary adenocarcinoma, NOS (8260/3); or papillary carcinoma, follicular variant (8340/3). We limited our study to patients with these 3 histology codes because grouping less-common subtypes could both bias our results and lead to a misrepresentation of our conclusions, if applied to the other subtypes. The subtypes of papillary carcinoma, NOS and papillary adenocarcinoma, NOS were combined to form the group referred to as conventional PTC (CPTC). Both are variants of classical PTC, characterized by a papillary architecture with the nuclear features of enlargement, crowding, clearing, and grooves.[3,9] Patients were included if they were >18 years of age, had a tumor size of 1.0–3.9 cm, clinical stage N0 or Nx, and had a lobectomy or total thyroidectomy. Patients were excluded if they had more than 1 primary malignancy, distant metastases, received chemotherapy, unknown radiation history, unknown multifocality, or unknown extrathyroidal extension. Patients with distant metastases or receiving chemotherapy were excluded to limit the analysis only to patients who underwent surgery for curative intent.

Patient variables analyzed were age at diagnosis (grouped by <45, 45–64, and ≥65), sex, race, insurance status, annual income, tumor size, nodal status, extent of disease, tumor multifocality, surgical margins, Charlson-Deyo comorbidity index (CCI), radiation treatment limited to radioactive iodine (RAI), hospital volume, and facility type.

Patients were further categorized into groups based on whether they had CPTC or FVPTC and received either lobectomy or total thyroidectomy. Lobectomy patients included those who underwent lobectomy with or without isthmusectomy. Total thyroidectomy patients included those who underwent total, near-total, or subtotal thyroidectomy. Patients who had a lobectomy followed by completion thyroidectomy were included in the total thyroidectomy group. Overall survival analyses were performed for subgroups defined by tumor sizes 1.0 to 3.9 cm, 1.0 to 1.9 cm, and 2.0 to 3.9 cm.

Patient characteristics were reported using means and frequencies. Descriptive data were compared across groups using the independent samples *t* test and Pearson χ^2 test. A Kaplan-Meier analysis was performed and survival distributions were compared across groups using the log-rank test. A multivariable Cox proportional hazards model stratified by tumor size was used to determine whether the extent of surgery contributed to an overall survival advantage for the 2 groups of CPTC and FVPTC. The model adjusted for the effects of patient demographics and clinical factors, including age at diagnosis, sex, race, insurance status and type, annual income level, nodal status, extrathyroidal extension, multifocality, surgical margins, CCI score, RAI, tumor size, hospital volume, and facility type because all were significant on univariable analysis.

Results

In the study, 126,416 patients with thyroid cancer were identified in the 2013 NCDB participant user file from 2004–2008. Of those, 33,816 patients met the inclusion criteria and had either CPTC or FVPTC with tumors between 1.0 and 3.9 cm (Fig 1).

Thyroid lobectomy was performed in 2,835 patients and total thyroidectomy in 30,981 patients. Of the 22,899 patients diagnosed with CPTC, 21,589 (94.3%) received a total thyroidectomy and 1,310 (5.7%) received a lobectomy (Table 1). In the study, 57.6% of CPTC patients had tumors 1.0 to 1.9 cm and 42.4% had tumors 2.0 to 3.9 cm. Compared to total thyroidectomy patients, patients who had lobectomies did not differ in mean tumor size (18.76 mm vs 18.90 mm, $P=.51$), sex (80.8% female vs 78.7% female, $P=.08$), or CCI score ($P=.09$). There was no difference in the mean follow-up time for CPTC patients undergoing a total thyroidectomy (6.49 ± 2.13 years) versus lobectomy (6.48 ± 2.33, $P=.92$). Lobectomy patients with CPTC were older (mean 47.01 years vs 45.39 years, $P<.001$), had a lower income (29.5% < $36,000 per year vs 21.2%, $P<.001$), and were more likely to be black (6.6% vs 4.6%, $P=.002$). They were less likely to have private insurance (72.5% vs 79.8%), be node positive (10.0% vs 27.2%), have extrathyroidal extension (9.5% vs 20.1%), exhibit multifocality (26.6% vs 42.5%), have positive surgical margins (7.9% vs 14.7%), receive RAI (40.7% vs 70.6%), be treated at a high volume center (23.3% vs 33.2%), or be treated at an academic or research center (21.9% vs 24.7%; all $P<.001$).

Of the 10,917 patients diagnosed with FVPTC, 9,392 (86.0%) had a total thyroidectomy and 1,310 (14.0%) had a lobectomy. FVPTC patients with tumors measuring 1.0 to 1.9 cm accounted for 51.2% and 48.8% had tumors 2.0 to 3.9 cm (Table 2). Compared to FVPTC patients undergoing total thyroidectomy, FVPTC patients undergoing lobectomies did not differ in mean tumor size (20.25 mm vs 19.94 mm), sex (80.3% female vs 81.7% female), race, insurance status, income, or CCI. The mean follow-up time for FVPTC patients undergoing a total thyroidectomy was 6.55 ± 2.17 years; for those undergoing a lobectomy, mean age was 6.43 ± 2.31 years ($P=.91$). Patients with lobectomies were younger (mean 46.67 years vs 47.67 years) and less likely to be node positive (2.4% vs 10.5%), have extrathyroidal extension (4.5% vs 11.2%), exhibit tumor multifocality (31.3% vs 45.1%), have positive surgical margins (4.5% vs 9.9%), undergo radiation treatment (46.1% vs 70.9%), be treated at a high-volume center (24.8% vs 33.2%), or be treated at an academic or research center (24.5% vs 27.5%; all $P<.001$).

Unadjusted overall survival was better for CPTC patients who received a total thyroidectomy compared to those who had a lobectomy ($\chi^2=5.74$, $P=.0166$; Fig 2, *A*). On stratification based on tumor size, this survival difference was most pronounced among CPTC patients with tumors 2.0 to 3.9 cm ($\chi^2=6.95$, $P=.0084$; Fig 3). Unadjusted overall survival was the same for FVPTC patients who received a total thyroidectomy compared to those who had a lobectomy ($\chi^2=0.66$, $P=.4165$; Fig 2, *B*). Stratification based on tumor size did not show differences in survival for FVPTC patients with 1.0- to 1.9-cm tumors ($P=.4431$) or 2.0- to 3.9-cm tumors ($P=.6986$).

Table 1
Characteristics for conventional PTC patients 2004–2008.

Variables	All ($n = 22,899$)		Total thyroidectomy ($n = 21,589$)		Lobectomy ($n = 1,310$)		*P* value
	Mean	SD	Mean	SD	Mean	SD	
Age, yr	45.48	14.19	45.39	14.13	47.01	15.04	.0001
Tumor size, mm	18.89	7.31	18.90	7.30	18.76	7.40	.5060
Follow-up time, yr	6.49	2.14	6.49	2.13	6.48	2.33	.9174
	n	%	*n*	%	*n*	%	
Age							
<45	11,446	50.0%	10,853	50.3%	593	45.3%	
45–64	9,012	39.4%	8,474	39.3%	538	41.1%	<.0001
≥65	2,441	10.7%	2,262	10.5%	179	13.7%	
Sex							
Male	4,844	21.2%	4,592	21.3%	252	19.2%	.0801
Female	18,055	78.8%	16,997	78.7%	1,058	80.8%	
Race							
White	19,250	84.1%	18,143	84.0%	1,107	84.5%	
Black	1,072	4.7%	986	4.6%	86	6.6%	
Hispanic	604	2.6%	579	2.7%	25	1.9%	.0023
Asian/Pacific Islander	1,189	5.2%	1,138	5.3%	51	3.9%	
Native American	57	0.2%	53	0.2%	4	0.3%	
Other/Unknown	727	3.2%	690	3.2%	37	2.8%	
Insurance							
None	535	2.3%	489	2.3%	46	3.5%	
Private	18,183	79.4%	17,233	79.8%	950	72.5%	
Medicaid	972	4.2%	903	4.2%	69	5.3%	<.0001
Medicare	2,595	11.3%	2,393	11.1%	202	15.4%	
Other	614	2.7%	571	2.6%	43	3.3%	
Income							
<$36,000	4,963	21.7%	4,576	21.2%	387	29.5%	
≥$36,000	17,016	74.3%	16,137	74.7%	879	67.1%	<.0001
Missing	920	4.0%	876	4.1%	44	3.4%	
Nodes status							
Negative	7,217	31.5%	6,879	31.9%	338	25.8%	
Positive	6,014	26.3%	5,883	27.2%	131	10.0%	<.0001
Nx	9,668	42.2%	8,827	40.9%	841	64.2%	
Extent of disease							
No extension	18,442	80.5%	17,256	79.9%	1,186	90.5%	<.0001
Extrathyroidal extension	4,457	19.5%	4,333	20.1%	124	9.5%	
Multifocality							
No	13,364	58.4%	12,403	57.5%	961	73.4%	<.0001
Yes	9,535	41.6%	9,186	42.5%	349	26.6%	
Surgical margins							
Negative	18,702	81.7%	17,567	81.4%	1,135	86.6%	
Positive	3,284	14.3%	3,181	14.7%	103	7.9%	<.0001
Unknown	913	4.0%	841	3.9%	72	5.5%	
Charlson Deyo Score							
0	20,188	88.2%	19,052	88.2%	1,136	86.7%	
1	2,352	10.3%	2,207	10.2%	145	11.1%	.0868
2+	359	1.6%	330	1.5%	29	2.2%	
Radiation treatment							
No	7,115	31.1%	6,338	29.4%	777	59.3%	<.0001
Yes	15,784	68.9%	15,251	70.6%	533	40.7%	
Tumor size							
1.0–1.9 cm	13,182	57.6%	12,426	57.6%	756	57.7%	.9135
2.0–3.9 cm	9,717	42.4%	9,163	42.4%	554	42.3%	
Hospital volume							
1–30	7,486	32.7%	6,854	31.7%	632	48.2%	
31–70	7,932	34.6%	7,559	35.0%	373	28.5%	<.0001
≥71	7,481	32.7%	7,176	33.2%	305	23.3%	
Facility type							
Community cancer program	897	3.9%	812	3.8%	85	6.5%	
Comprehensive community cancer Program	6,789	29.6%	6,356	29.4%	433	33.1%	
Academic/research program	5,620	24.5%	5,333	24.7%	287	21.9%	<.0001
Integrated network cancer program	1,179	5.1%	1,112	5.2%	67	5.1%	
Other specified types of cancer program	13	0.1%	13	0.1%	0	0.0%	
Missing	8,401	36.7%	7963	36.9%	438	33.4%	

However, when adjusting for patient and clinical factors, CPTC patients with tumor sizes grouped from 1.0 to 3.9 cm did not show a survival advantage with total thyroidectomy compared to lobectomy (hazard ratio [HR] 1.04; 95% confidence interval [CI], 0.80–1.36, $P = .75$; Table 3). Adjusted overall survival for FVPTC patients with tumor sizes grouped from 1.0 to 3.9 cm also did not show a survival advantage with total thyroidectomy compared to lobectomy (HR 0.89; CI, 0.67–1.20, $P = .46$; Table 3).

On adjusted analysis for CPTC and FVPTC further stratified by tumor size groups 1.0 to 1.9 cm and 2.0 to 3.9 cm, CPTC patients

Table 2
Characteristics for FVPTC patients 2004–2008.

Variables	All ($n = 10,917$)		Total thyroidectomy ($n = 9,392$)		Lobectomy ($n = 1,525$)		*P* value
	Mean	SD	Mean	SD	Mean	SD	
Age, yr	47.53	13.99	47.67	13.85	46.67	14.76	.0128
Tumor size, mm	19.99	7.76	19.94	7.77	20.25	7.72	.1552
Follow-up time, yr	6.53	2.19	6.55	2.17	6.43	2.31	.0672
	n	%	*n*	%	*n*	%	
Age							
<45	4,697	43.0%	3,981	42.4%	716	47.0%	
45–64	4,840	44.3%	4,240	45.1%	600	39.3%	.0001
≥65	1,380	12.6%	1,171	12.5%	209	13.7%	
Sex							
Male	2,020	18.5%	1,719	18.3%	301	19.7%	.1808
Female	8,897	81.5%	7,673	81.7%	1,224	80.3%	
Race							
White	9,124	83.6%	7860	83.7%	1,264	82.9%	
Black	856	7.8%	724	7.7%	132	8.7%	
Hispanic	220	2.0%	197	2.1%	23	1.5%	.1216
Asian/Pacific Islander	395	3.6%	327	3.5%	68	4.5%	
Native American	21	0.2%	18	0.2%	3	0.2%	
Other/Unknown	301	2.8%	266	2.8%	35	2.3%	
Insurance							
None	241	2.2%	204	2.2%	37	2.4%	
Private	8,423	77.2%	7,275	77.5%	1,148	75.3%	
Medicaid	546	5.0%	454	4.8%	92	6.0%	.1771
Medicare	1,468	13.4%	1,250	13.3%	218	14.3%	
Other	239	2.2%	209	2.2%	30	2.0%	
Income							
<$36,000	2,578	23.6%	2,201	23.4%	377	24.7%	
≥$36,000	7,945	72.8%	6,837	72.8%	1,108	72.7%	.0579
Missing	394	3.6%	354	3.8%	40	2.6%	
Nodes status							
Negative	3,593	32.9%	3,206	34.1%	387	25.4%	
Positive	1,026	9.4%	990	10.5%	36	2.4%	<.0001
Nx	6,298	57.7%	5,196	55.3%	1,102	72.3%	
Extent of disease							
No Extension	9,798	89.7%	8,341	88.8%	1,457	95.5%	<.0001
Extrathyroidal extension	1,119	10.3%	1,051	11.2%	68	4.5%	
Multifocality							
No	6,207	56.9%	5,160	54.9%	1,047	68.7%	<.0001
Yes	4,710	43.1%	4,232	45.1%	478	31.3%	
Surgical margins							
Negative	9,539	87.4%	8,140	86.7%	1,399	91.7%	
Positive	995	9.1%	927	9.9%	68	4.5%	<.0001
Unknown	383	3.5%	325	3.5%	58	3.8%	
Charlson Deyo Score							
0	9,499	87.0%	8,168	87.0%	1,331	87.3%	
1	1,220	11.2%	1,058	11.3%	162	10.6%	.5245
≥2	198	1.8%	166	1.8%	32	2.1%	
Radiation treatment							
No	3,559	32.6%	2,737	29.1%	822	53.9%	<.0001
Yes	7,358	67.4%	6,655	70.9%	703	46.1%	
Tumor size							
1.0–1.9 cm	5,593	51.2%	4,846	51.6%	747	49.0%	.0582
2.0–3.9 cm	5,324	48.8%	4,546	48.4%	778	51.0%	
Hospital volume							
1–30	3,815	34.9%	3,144	33.5%	671	44.0%	
31–70	3,609	33.1%	3,133	33.4%	476	31.2%	<.0001
≥71	3,493	32.0%	3,115	33.2%	378	24.8%	
Facility type							
Community cancer program	610	5.6%	492	5.2%	118	7.7%	
Comprehensive community cancer program	3,479	31.9%	3,017	32.1%	462	30.3%	
Academic/Research program	2,955	27.1%	2,582	27.5%	373	24.5%	<.0001
Integrated network cancer program	567	5.2%	504	5.4%	63	4.1%	
Other specified types of cancer program	3	0.0%	3	0.0%	0	0.0%	
Missing	3,303	30.3%	2,794	29.7%	509	33.4%	
Facility type							
Community cancer program	610	5.6%	492	5.2%	118	7.7%	
Comprehensive community cancer program	3,479	31.9%	3,017	32.1%	462	30.3%	
Academic/Research program	2,955	27.1%	2,582	27.5%	373	24.5%	<.0001
Integrated network cancer program	567	5.2%	504	5.4%	63	4.1%	
Other specified types of cancer program	3	0.0%	3	0.0%	0	0.0%	
Missing	3,303	30.3%	2,794	29.7%	509	33.4%	

Table 3
Multivariable Cox regression models, size 1.0–3.9 cm.

Variables	Conventional PTC				FVPTC			
	95% CI				95% CI			*P* value
	HR	Low	High	*P* value	HR	Low	High	
Age								
<45 (reference)								
45–64	2.61	1.82	3.76	<.001	2.14	1.27	3.61	.0041
≥65	9.81	6.53	14.75	<.001	4.97	2.77	8.93	<.001
Sex								
Male	1.46	1.26	1.70	<.0001	1.85	1.49	2.29	<.0001
Female (reference)								
Race								
White (reference)								
Black	1.17	0.87	1.57	.2907	1.11	0.79	1.58	.5453
Hispanic	0.63	0.38	1.04	.0710	0.26	0.07	1.07	.0613
Asian/Pacific Islander	0.69	0.46	1.02	.0633	0.41	0.15	1.12	.0817
Native American	0.65	0.09	4.66	.6710				
Other/Unknown	0.73	0.42	1.26	.2543	0.83	0.39	1.77	.6368
Insurance								
Private (reference)								
None	2.16	1.37	3.42	.0010	1.16	0.43	3.17	.7681
Medicaid	2.01	1.39	2.90	.0002	2.70	1.71	4.26	<.0001
Medicare	1.80	1.42	2.28	<.0001	3.11	2.18	4.44	<.0001
Other	1.94	1.30	2.92	.0013	1.43	0.63	3.26	.3900
Income								
<$36,000	1.29	1.11	1.50	.0011	1.18	0.94	1.48	.1499
≥$36,000 (reference)								
Nodes status								
Negative (reference)								
Positive	1.92	1.57	2.36	<.0001	1.03	0.67	1.57	.9037
Nx	1.31	1.09	1.58	.0036	1.13	0.89	1.44	.3289
Extent of disease								
No extension (reference)								
Extrathyroidal extension	1.56	1.32	1.85	<.0001	1.06	0.76	1.48	.7392
Multifocality								
No (reference)								
Yes	0.94	0.82	1.09	.4260	0.98	0.80	1.21	.8489
Surgical margins								
Negative (reference)								
Positive	1.11	0.92	1.34	.2635	1.33	0.95	1.86	.1027
Charlson Deyo Score								
0 (reference)								
1	1.72	1.44	2.05	<.0001	1.79	1.39	2.29	<.0001
≥2	4.11	3.23	5.23	<.0001	3.52	2.48	5.00	<.0001
Radiation treatment								
No	1.55	1.33	1.80	<.0001	1.39	1.13	1.72	.0023
Yes (reference)								
Tumor size								
1.0–1.9 cm (reference)								
2.0–3.9 cm	1.13	0.99	1.31	.0812	1.15	0.94	1.41	.1673
Hospital volume								
1–30	1.13	0.92	1.38	.2390	1.13	0.83	1.54	.4344
31–70	0.94	0.78	1.13	.5098	0.98	0.74	1.30	.8872
≥71 (reference)								
Facility type								
Community cancer program (reference)								
Comprehensive community cancer program	1.01	0.76	1.35	.9399	0.87	0.59	1.27	.4664
Academic/Research program	1.09	0.80	1.50	.5775	0.92	0.60	1.41	.7032
Integrated network cancer program	1.15	0.77	1.72	.4886	1.07	0.62	1.85	.8026
Other specified types of cancer program	1.68	0.23	12.37	.6097				
Missing	0.84	0.51	1.39	.4964	0.66	0.32	1.34	.2482

with tumors 1.0 to 1.9 cm did not show an overall survival advantage with total thyroidectomy compared to lobectomy (HR 0.76; CI, 0.51–1.12, $P = .1596$), but patients with tumors 2.0 to 3.9 cm did (HR 1.53; CI 1.06–2.19, $P = .0226$; Table 4). Factors independently associated with worse survival in patients with tumors 2.0 to 3.9 cm were older age, male sex, black race, not having private insurance, positive or Nx nodal status, extrathyroidal extension, unknown surgical margins, higher CCI score, and not receiving RAI. FVPTC patients did not show a survival advantage with total thyroidectomy over lobectomy in the adjusted model for tumors 1.0 to 1.9 cm (HR 0.85; CI, 0.55–1.30, $P = .4489$) or 2.0 to 3.9 cm (HR 0.97; CI, 0.65–1.45, $P = .8848$; Table 5).

Sensitivity analysis

To examine the effect of patients who were potentially mistreated by having received a lobectomy and RAI, thereby confounding previous survival analyses, a subset analysis was performed excluding these patients. In total, 533 lobectomy patients with CPTC and 703 lobectomy patients with FVPTC were excluded. Adjusted

Table 4
Multivariable Cox regression models by tumor size, conventional PTC patients.

Variables	1.0–1.9 cm				2.0–3.9 cm			
	HR	95% CI		*P* value	HR	95% CI		*P* value
		Low	High			Low	High	
Age								
<45 (reference)								
45–64	1.77	1.15	2.74	.0097	4.81	2.40	9.65	<.0001
≥65	7.19	4.34	11.91	<.0001	16.56	7.92	34.65	<.0001
Sex								
Male	1.68	1.36	2.07	<.0001	1.29	1.04	1.60	.0205
Female (reference)								
Race								
White (reference)								
Black	0.70	0.43	1.14	.1532	1.78	1.24	2.57	.0019
Hispanic	0.62	0.30	1.29	.1976	0.68	0.35	1.31	.2513
Asian/Pacific Islander	0.74	0.43	1.25	.2598	0.72	0.41	1.28	.2696
Native American	2.83	0.56	14.25	.2072	0.47	0.03	7.62	.5937
Other/Unknown	0.65	0.30	1.43	.2847	0.90	0.43	1.89	.7874
Insurance								
Private (reference)								
None	2.62	1.40	4.93	.0027	1.94	1.03	3.68	.0416
Medicaid	2.20	1.31	3.69	.0028	1.94	1.16	3.25	.0116
Medicare	1.66	1.18	2.32	.0034	1.99	1.42	2.78	<.0001
Other	2.34	1.38	3.98	.0016	1.66	0.90	3.07	.1031
Income								
<$36,000	1.47	1.19	1.81	.0004	1.15	0.92	1.43	.2097
≥$36,000 (reference)								
Nodes status								
Negative (reference)								
Positive	1.74	1.31	2.31	.0001	2.12	1.58	2.86	<.0001
Nx	1.28	1.00	1.64	.0498	1.38	1.04	1.83	.0239
Extent of disease								
No extension (reference)								
Extrathyroidal Extension	1.44	1.12	1.85	.0043	1.73	1.38	2.17	<.0001
Multifocality								
No (reference)								
Yes	0.91	0.74	1.11	.3528	1.01	0.82	1.23	.9526
Surgical margins								
Negative (reference)								
Positive	1.13	0.85	1.50	.3949	1.13	0.88	1.44	.3491
Unknown	1.24	0.80	1.93	.3458	1.83	1.23	2.72	.0030
Charlson Deyo Score								
0 (reference)								
1	2.00	1.57	2.54	<.0001	1.49	1.15	1.93	.0024
≥2	4.72	3.36	6.63	<.0001	3.71	2.63	5.24	<.0001
Radiation treatment								
No	1.60	1.30	1.96	<.0001	1.53	1.23	1.91	.0001
Yes (reference)								
Hospital volume								
1–30	1.26	0.95	1.68	.1092	1.02	0.76	1.37	.8885
31–70	0.94	0.72	1.23	.6536	0.94	0.72	1.23	.6495
≥71 (reference)								
Facility type								
Community cancer program (reference)								
Comprehensive community cancer program	0.84	0.57	1.25	.3978	1.26	0.81	1.97	.3020
Academic/Research program	1.03	0.67	1.58	.9027	1.23	0.76	1.97	.3965
Integrated network cancer program	1.09	0.63	1.88	.7572	1.42	0.78	2.58	.2534
Other specified types of cancer program	10.33	1.84	58.00	0.0080	1.00	0.06	17.55	.9987
Missing	0.65	0.35	1.21	0.1732	1.42	0.59	3.38	.4318
Surgery								
Total thyroidectomy (reference)								
Lobectomy	0.76	0.51	1.12	0.1596	1.53	1.06	2.19	.0226

overall survival for CPTC and FVPTC was re-examined, and the results were unchanged. CPTC patients who received a total thyroidectomy for tumors 1.0 to 1.9 cm still did not show an overall survival advantage (HR 1.07; CI 0.70–1.65, $P = .7554$), whereas patients with tumors 2.0 to 3.9 cm did (HR 1.68; CI, 1.05–2.70, $P = .0323$; Supplementary Table S1). FVPTC patients who received a total thyroidectomy did not show an overall survival advantage for tumors 1.0 to 1.9 cm (HR 1.09; CI 0.69–1.71, $P = .7243$) or 2.0 to 3.9 cm (HR 1.54; CI, 0.97–2.42, $P = .0647$; Supplementary Table S2).

Discussion

The decision to perform a lobectomy or total thyroidectomy for well-differentiated thyroid cancer has been debated for years. One of the biggest changes from the 2009 to 2015 ATA guidelines was the statement that lobectomy can be considered for clinically N0 thyroid carcinomas with tumor size 1.0 to 3.9 cm. The arguments favoring lobectomy typically cite the excellent prognosis of PTC, the indolent nature of the disease, and the increased risk of po-

Table 5
Multivariable Cox regression models by tumor size, FVPTC patients.

Variables	1.0–1.9 cm				2.0–3.9 cm			
	HR	95% CI		*P* value	HR	95% CI		*P* value
		Low	High			Low	High	
Age								
<45 (reference)								
45–64	2.01	0.94	4.26	.0705	2.08	1.05	4.11	.0354
≥65	7.88	3.40	18.27	<.0001	2.95	1.38	6.28	.0051
Sex								
Male	1.75	1.27	2.39	.0005	2.02	1.49	2.74	<.0001
Female (reference)								
Race								
White (reference)								
Black	1.20	0.73	1.97	.4638	1.13	0.69	1.84	.6313
Hispanic	0.44	0.09	2.25	.3239	0.35	0.07	1.77	.2053
Asian/Pacific Islander	0.29	0.06	1.45	.1302	0.78	0.27	2.28	.6492
Native American	3.02	0.18	50.11	.4400	0.64	0.04	11.68	.7660
Other/Unknown	1.09	0.46	2.60	.8465	0.71	0.20	2.51	.5941
Insurance								
Private (reference)								
None	1.77	0.50	6.32	.3798	1.17	0.33	4.20	.8058
Medicaid	2.64	1.32	5.26	.0060	2.80	1.53	5.10	.0008
Medicare	1.98	1.21	3.25	.0068	4.77	2.95	7.71	<.0001
Other	1.94	0.74	5.10	.1801	1.38	0.39	4.92	.6182
Income								
<$36,000	0.85	0.60	1.20	.3570	1.63	1.19	2.22	.0022
≥$36,000 (reference)								
Nodes status								
Negative (reference)								
Positive	1.19	0.65	2.18	.5636	0.88	0.48	1.61	.6808
Nx	1.17	0.82	1.65	.3934	1.02	0.73	1.44	.9005
Extent of disease								
No extension (reference)								
Extrathyroidal extension	1.11	0.69	1.79	.6564	1.01	0.63	1.63	.9564
Multifocality								
No (reference)								
Yes	0.94	0.70	1.26	.6807	0.97	0.72	1.31	.8477
Surgical margins								
Negative (reference)								
Positive	1.04	0.63	1.73	.8763	1.73	1.10	2.74	.0185
Unknown	0.59	0.23	1.51	.2696	0.56	0.19	1.62	.2816
Charlson Deyo Score								
0 (reference)								
1	1.74	1.23	2.47	.0018	1.88	1.31	2.68	.0005
≥2	2.58	1.50	4.41	.0006	4.26	2.66	6.84	<.0001
Radiation treatment								
No	1.18	0.87	1.62	.2908	1.67	1.24	2.25	.0007
Yes (reference)								
Hospital volume								
1–30	1.43	0.91	2.25	.1210	0.90	0.58	1.38	.6202
31–70	1.16	0.77	1.73	.4834	0.89	0.60	1.32	.5552
≥71 (reference)								
Facility type								
Community cancer program (reference)								
Comprehensive community cancer program	0.76	0.44	1.30	.3111	0.99	0.58	1.71	.9787
Academic/Research program	0.95	0.52	1.74	.8587	0.90	0.49	1.66	.7440
Integrated network cancer program	1.25	0.60	2.61	.5555	0.96	0.42	2.23	.9304
Other specified types of cancer program	9.80	0.52	185.55	.1283				
Missing	0.61	0.21	1.76	.3617	0.68	0.27	1.72	.4108
Surgery								
Total thyroidectomy (reference)								
Lobectomy	0.85	0.55	1.30	.4489	0.97	0.65	1.45	.8848

tential complications as a reason to avoid total thyroidectomy in the absence of a clear survival advantage.[10] On the other hand, total thyroidectomy eliminates the potential for disease in the contralateral lobe given the multifocal nature of PTC, enables the use of RAI for postoperative treatment, and facilitates the use of serum thyroglobulin as a marker of residual disease and recurrence.[11-13] The 2009 ATA guidelines provided a strong recommendation stating tumor size alone was adequate in determining the optimal extent of thyroid surgery for PTC, as Bilimoria et al had shown a survival advantage in patients who underwent a total thyroidectomy for tumors >1 cm.[6] Adam et al in their analysis of 61,775 patients in the NCDB from 1998–2006 performed an updated study taking into account additional variables in the NCDB and showed no survival advantage for PTC patients who underwent total thyroidectomy compared to lobectomy for tumors 1.0 to 4.0 cm, 1.0 to 2.0 cm, and 2.1 to 4.0 cm.[5] The use of a more recent cohort of patients in the face of increased quality measures by the NCDB, as well as the inclusion of factors that were not available at the

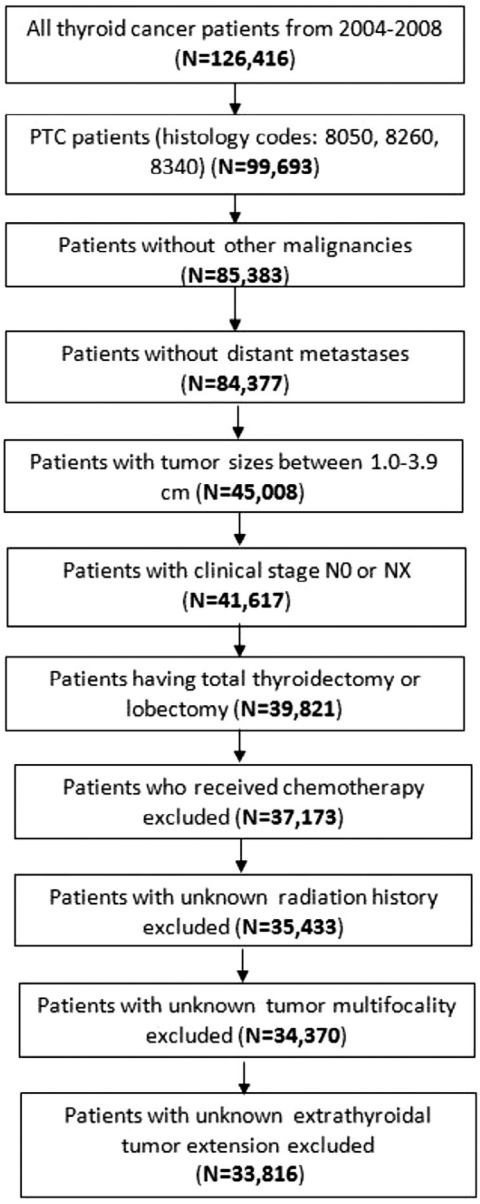


Fig. 1. Patient selection map.

time of the Bilimoria publication (i.e., comorbidity, multifocality, extrathyroidal extension, and completeness of surgical resection) were cited as the cause for the difference between the 2 studies. We hypothesized that the conflicting outcomes were due to the different histologic subtypes in the 2 studies. Not all patients in the Adam et al study had data on comorbidity, multifocality, and extrathyroidal extension, which may account for the differences in these 2 studies.

Of note is the evolving pattern of surgical treatment of papillary thyroid cancer. In the Bilimoria et al study of the 1985–1998 NCDB cohort, 17% of patients had lobectomy; the Adams et al study of 1998–2006 NCDB cohort saw the rate of lobectomies decrease to 11%. We believe the further decrease of lobectomy to 8% in our study is a response to the 2009 ATA Guidelines recommending total thyroidectomy for all papillary cancers >1 cm. In our own review of the NCDB database, we observed that the rate of lobectomy decreased from 14% for patients treated in 1998 to 8% for those treated in 2007. This change in treatment should not affect our results since we are still evaluating substantial populations in each group.

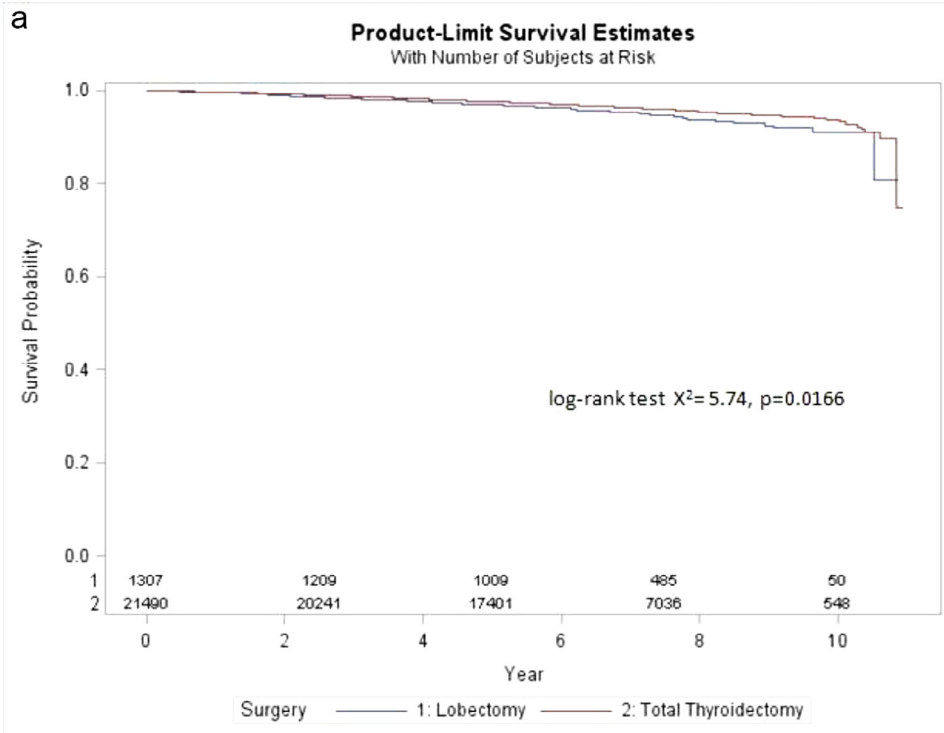
In our study as well as studies by others,[3,14] FVPTC patients were on average older and yet still demonstrated a more favorable prognosis when compared to CPTC patients. Age is a critical factor in determining prognosis of PTC. Patients with CPTC treated with total thyroidectomy were younger than those treated with lobectomy (45.4 vs 47.0 years, respectively). Although this is a small difference, it is highly significant and likely reflects an effort by clinicians to choose the less aggressive surgery for older patients with more comorbidities. The multivariable Cox regression model with age as a reference should control for the effect of this variable.

FVPTC in multiple studies has been shown to have a different biologic behavior than CPTC.[3,14-18] In addition, the incidence of FVPTC has been increasing in recent years and is the most common subtype of PTC.[14-18] In a single-institution, retrospective study from the Northern Israel Oncology Center published in Cancer, FVPTC accounted for 41% of PTC cases.[19] These findings emphasize the importance of histologic variant–based treatment recommendations for CPTC and FVPTC, similar to those for the tall cell variant of PTC, which is a well-established variant for which more aggressive treatment is recommended and accepted.[20] Our findings of no difference between overall survival for FVPTC on both unadjusted and adjusted analysis based on tumor size supports these previously published data.

Our results show CPTC patients with 2.0- to 3.9-cm tumors who received lobectomies had worse overall survival, which suggests both histology and treatment choice play important roles in affecting outcome. These findings shed light on the contradictory outcomes of the Adams et al and Bilimoria et al studies. Differences in the inclusion criteria for the 2 studies have a major impact on the conclusions. The relatively more aggressive nature of CPTC versus FVPTC, established by Shi et al from a clinicopathologic standpoint, is corroborated by our results on overall survival using the NCDB.[3] While the mean follow-up in the Shi et al study is relatively short for PTC at 37 months, a smaller but more long-term study by Prassler et al showed that FVPTC still had superior disease-specific survival compared to CPTC (100% vs 93.1%) at 10-year follow-up despite having similar recurrence rates of 8% vs 10%.[21] Our results support the conclusion that surgeons should consider a total thyroidectomy for patients with CPTC tumors ≥2.0 cm, because fine-needle aspiration alone cannot determine the histologic subtype of PTC preoperatively. Preoperative molecular testing fine-needle aspiration of samples has shown promise for differentiating FVPTC from CPTC; as these assays improve, they may become reliable preoperative determinants of the histologic subtype and guide surgical decision making.[22-24]

Limitations

Our study is limited by the relatively short follow-up period, especially in the face of the overall excellent prognosis of thyroid cancer. In addition, it is limited by factors inherent to the use of the NCDB, including the accuracy of the database and lack of disease recurrence data. The NCDB also does not contain disease specific survival data, which would be important given the overall favorable prognosis for PTC. However, the reporting of disease specific mortality can be inconsistent and is subject to bias in reporting, and overall survival offers a more objective meaure.[25] Another limitation is the inability to retrospectively review the pathology as there has been further reclassification of encapsulated FVPTC as noninvasive follicular tumor with papillary like features.[26] It is possible that some of the FVPTC tumors would now be classified as noninvasive follicular tumor with papillary like features, and patients with these lesions should have much better survival. We believe they would be equally distributed among the FVPTC patients



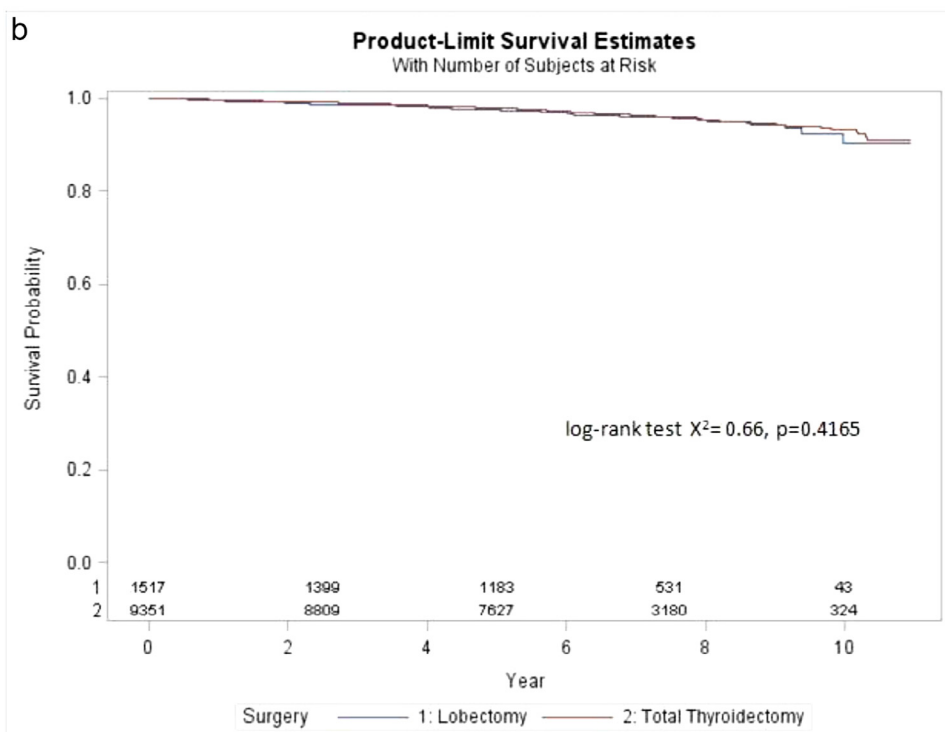


Fig. 2. (*A*) Patient KM survival curve by surgery: conventional PTC patients ($N = 22{,}899$). (*B*) Patient KM survival curve by surgery: FVPTC patients ($N = 10{,}917$).

having total thyroidectomy compared with those having lobectomy. Current data in the NCDB cannot clarify this issue. Its strengths derive from the large number of patients, the use of patients after increased data quality measures were instituted, and the use of a contemporary cohort of patients in whom all variables of question were available. In addition, our efforts to eliminate the bias of inappropriately treated patients having lobectomy and receiving RAI did not change our conclusion (Tables 1 and 2).

The histologic subtype of PTC, in conjunction with size and extent of surgery, plays an important role in the overall survival of patients with PTC. Total thyroidectomy may be associated with an improved overall survival in patients with conventional PTC for tumors 2.0 to 3.9 cm and should be considered as the initial surgical procedure. Lobectomy for FVPTC tumors ranging from 1.0 to 3.9 cm is sufficient and not associated with worse overall survival.

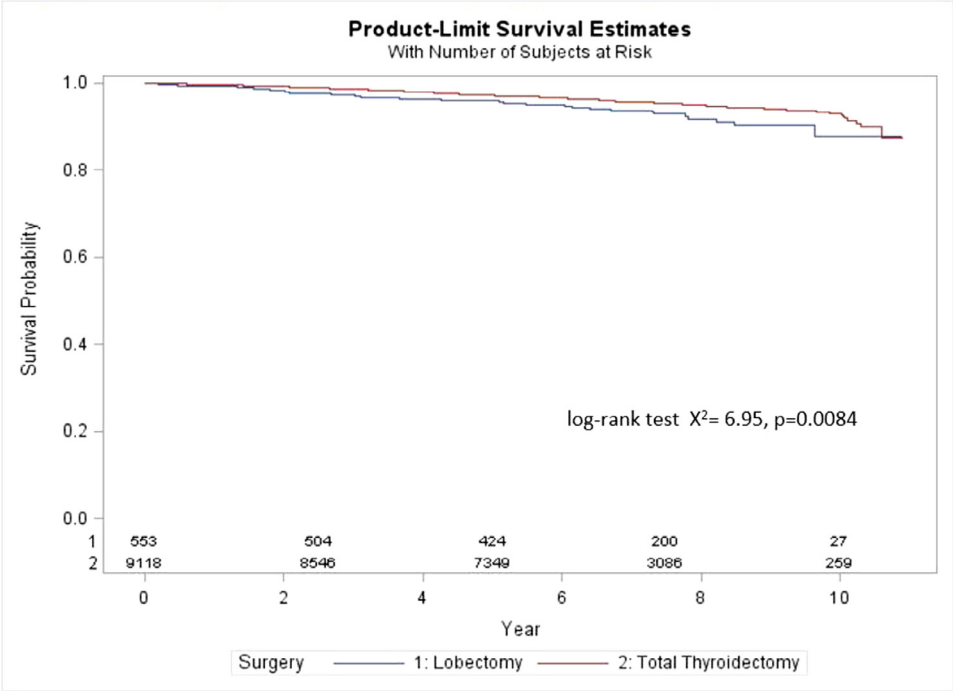


Fig. 3. Patient KM survival curve by surgery: conventional PTC patients (tumor size 2.0 to 3.9 cm).

Supplementary data

Supplementary data related to this article can be found online at https://doi.org/10.1016/j.surg.2017.12.026.

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